

Alarms Dataset

Data Specification

June 2026

Overview

This dataset contains **18,702 alarm events** from building management and fire safety systems at Valmet industrial sites. Data covers **January 2025 – May 2026** (~17 months). Includes HVAC alarms, fire alarms, fire equipment faults, line faults, and informational events.

Each row is a single alarm event received by Luotea's alarm monitoring center. Events are generated by hardware devices and logged with timestamps, priority, and description.

Language: EVENT_DESCRIPTION is in **Finnish**. No anonymization required.

File Details

Property	Value
Filename	alarms.csv
Format	CSV (UTF-8 with BOM)
Rows	18,702
Columns	25
Date range	2025-01-20 – 2026-05-25

Sites

Customer	Site	Events
Valmet Flow Control Oy	Toimistotalo	13,310
Valmet Flow Control Oy	Venttiilitehdas	3,617
Valmet Technologies Oy	Lentokentänkatu 11	1,536
Valmet Flow Control Oy	STD tehdas	236
Valmet Flow Control Oy	Warehouse	3

Column Reference

Identifiers

Column	Type	Null?	Description
ALERT_EVENT_ID	Integer	No	Primary key. Unique per row.

Column	Type	Null?	Description
ALERT_ID	Integer	No	Alert ID. One alert can produce multiple rows across locations.
CUSTOMER_NO	Integer	No	Customer number in Luotea's ERP. Shared key with work orders.
CUSTOMER_NAME	Text	No	Customer company name
CUSTOMER_SITE_NO	Integer	No	Site identifier
CUSTOMER_SITE_NAME	Text	No	Site name
SITE_ADDRESS	Text	No	Physical address
CONTRACT_NO	Integer	No	Contract number (5 unique)

Event Details

Column	Type	Null?	Description
EVENT_TIME	Datetime	No	When the alarm occurred. YYYY-MM-DD HH:MM:SS.000
EVENT_TYPE	Categorical	No	Always 'Hälytys' (Alarm) in this dataset
EVENT_DESCRIPTION	Text	No	Free-text alarm description in Finnish (1,694 unique)
EVENT_CLOSED_TIME	Datetime	Yes	When the alarm was closed/acknowledged

Hardware & Alarm Classification

Column	Type	Null?	Description
HWID	Text	No	Hardware device ID (11 unique devices)
HWID_TYPE	Categorical	No	Alerta (fire panel), K2 vastaanotin (BMS), Sähköpostihälytys (email)
LOOP	Text	Yes	Loop/circuit ID. Note: values have leading spaces.
ALERT_TYPE	Categorical	Yes	Alarm category (see table below)
Priority	Integer	No	Alarm priority (see table below)
LOG_ONLY	Categorical	Yes	'L' = log-only (33%). NULL = actionable.

ALERT_TYPE	Count	Description
LVIS	8,882	HVAC alarm
LINJAVIK	2,770	Line/communication fault
HÄLYT	2,579	General alarm
INFO	1,656	Informational event
LVIA	1,458	HVAC alarm variant
PALO	970	Fire alarm
PALOVIK	280	Fire equipment fault

Priority	Count	Description
1	970	Fire alarm (highest)
50	106	Medium
60	12,919	Standard HVAC/technical
77	3,050	Fire equipment / line fault
99	1,656	Informational / log-only

Work Order Link & Status

Column	Type	Null?	Description
WORKORDER_NO	Integer	Yes (96%)	Work order number. Links to WO_NO in work orders. Only 4% of alarms generate a work order.
LOCATION_ID	Integer	No	Location point identifier (20 unique)
LOCATION_TABLE_ID	Integer	No	Location table reference
IVA_NO	Text	No	Monitoring contract number (6 unique)
STATUS	Categorical	No	Always 'Valvonnassa' (Under monitoring)

Key Relationships

To Work Orders: WORKORDER_NO → WO_NO in work_orders_anonymized.csv (sparse: ~4% of alarms)

To Sites: CUSTOMER_NO + CUSTOMER_SITE_NO shared across all datasets

Notes for Participants

1. **One alert, many rows.** A single ALERT_ID can produce multiple rows — one per LOCATION_ID. Group by ALERT_ID for unique events.
2. **LOOP values have leading spaces** (e.g., ' 1'). Strip whitespace before analysis.
3. **96% of alarms have no work order.** The WORKORDER_NO link is sparse.
4. **Priority is not sequential.** Values: 1, 50, 60, 77, 99.
5. **EVENT_DESCRIPTION is in Finnish.** Contains device codes, temperatures, system identifiers.
6. **Encoding:** UTF-8 with BOM.